Frank Tinsley

Technical Artist (VR) · Real-Time Shader & Rendering Engineer (Unity, C#, GPU)

Portland, OR — open to Bay Area (relocation-ready) or remote

franktinsley@me.com · linkedin.com/in/franktinsley · franktinsley.github.io

SUMMARY

Technical Artist and real-time graphics engineer with 15+ years across Unity, C#, shaders, and GPU rendering — most recently contract Technical Artist at Meta (via Insight Global), working as part of a Meta team on Meta Avatars as it applies to real-time interactivity and rendering. Solo-built a production-grade SDF ray-marching 3D modeling engine in Swift + Metal — ~50,000 lines of Swift/Metal across ~650 commits — running in real time on iOS, iPadOS, and macOS. Comfortable on both sides of the art–engineering boundary: shader authoring, rendering-pipeline optimization, and cross-functional collaboration across art and engineering — prototyping fast, then hardening for real-time performance.

SKILLS

- **Real-time rendering & shaders:** shader authoring and optimization (HLSL/ShaderLab, Metal MSL), GPU compute, real-time rendering pipelines, graphics pipeline optimization, SDF / implicit surfaces, ray marching, procedural geometry, marching cubes
- **Engines & languages:** Unity, C#, Swift, SwiftUI, Metal, TypeScript/JavaScript, Go, Java
- **VR/AR/XR:** avatars, ARKit, Meta Spark AR, Lens Studio, mixed reality; cross-platform iOS/iPadOS/macOS and Android
- **Tooling:** file-format and pipeline tooling engineering (created the ARX AR interchange format); AI-accelerated development workflows

EXPERIENCE

Independent Software Developer — Self-Employed • Portland, OR · May 2023 – Present

Code and live demo available on request.

- Designed and built, solo, a production-grade real-time signed-distance-field (SDF) ray-marching 3D modeling engine and app in Swift + Metal (Feb–May 2026), running across iOS, iPadOS, and macOS.
- Authored the full shader stack — Metal compute, fragment, and vertex pipelines — for ray-marched implicit surfaces, smooth booleans and organic blending, and procedural geometry, optimized for interactive frame rates on mobile and desktop GPUs.
- Implemented marching-cubes mesh export (STL/OBJ), bridging the implicit-surface renderer into standard 3D content pipelines.
- Designed and implemented an FPGA re-implementation of the Konami System 573 arcade platform for MiSTer (source available on request) as a side project — ~22,000 lines of custom RTL in about one month, built through an AI-accelerated workflow and proven frame-accurate against the MAME reference emulator via automated frame-diff tooling on real FPGA hardware.

Technical Artist — Meta (contract via Insight Global) • Portland, OR (onsite) · Mar 2025 – Feb 2026

- Designed and built a visual in-editor tool for authoring mock events and controlling them at runtime, enabling full behavioral testing of the live avatar systems and continual evolution of Meta's internal live-avatars API.
- Worked across the art–engineering boundary on Meta Avatars — Meta's avatar system for VR and mixed-reality platforms — focusing on real-time interactivity and avatar rendering.

AR Subject Matter Expert — Meta (contract via IntraEdge) • Mar 2022 – May 2023

- Served as Meta's subject matter expert on Spark AR Studio and AR development more broadly; created hands-on activities and tutorials for an educational curriculum designed to train the next generation of AR creators.

Mixed Reality Developer — Microsoft • Remote · Sep 2021 – Mar 2022

Runtime Engineer — Camera IQ • Santa Monica, CA · Jan 2018 – Nov 2020

- Built key components of the Camera IQ platform's runtime and backend, starting in C# for the Unity runtime, then

JavaScript for additional AR runtimes.

- Implemented backend services first in Java (with Swift for UI automation), then in Go — while continuing to develop new runtime features in JavaScript — including a Go wrapper covering nearly the entire Spark AR Studio file format.
- Created ARX, an API-neutral AR experience file format for interchange between AR runtimes including Snapchat's Lens Studio and Facebook's Spark AR Studio.

Principal / **Contract Unity Developer — CodeStep LLC** (independent consultancy) • Tigard, OR · Nov 2009 – Jan 2018

- Delivered interactive experiences in Unity for mobile, XR, and branded retail installations, owning the entire creative and technical development process end to end.
- Built an interactive 3D installation for a major athletic shoe brand, exhibited at trade shows worldwide and in retail, plus original photo-booth software installed in retail locations.
- Developed the game "Fascination," purchased by GSN, then worked with GSN's team to improve it; built Unity game systems for the cross-platform mobile wellness game "Habit Monster."